

Lecture 4: Consumer Choice (Chapter 5)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, Chapter 5 — Consumer choice (§5.1 modelling choice · §5.2 preferences & utility · §5.3 budget constraints · §5.4 utility maximization · §5.5 discussion of the framework · §5.6 behavioral economics · §5.7 appendix: solving for the optimal basket with Lagrange). *Lecturer: Luigi Butera — deck "Lecture 4: Consumer choice."* (Numbering note: this is the course's 4th lecture but the textbook's Chapter 5; the deck's own "4.1–4.5" roadmap just re-numbers the chapter's sections.)

Theme of the lecture (Butera — from the deck). So far (Lectures 2–3) we took **demand curves as given** and pushed them around. Now we **build demand from the ground up**: *where do preferences come from, and how does a rational consumer choose under scarcity?* The deck's answer is a **three-stage machine** — **preferences → utility → budget constraint → constrained optimization** — that will (next chapter) *generate* the demand curve. The deck's five stops: **4.1 preferences · 4.2 utility · 4.3 budget constraints · 4.4 constrained choice · 4.5 behavioral economics.**

★ **The heart of the lecture:** a consumer **maximizes utility subject to a budget constraint**. We (1) describe what she **likes** with well-behaved **preferences** (complete + transitive), (2) quantify them with a **utility function** whose slope-of-indifference-curve is the **MRS** = how much of one good she'll trade for the other *and stay equally happy*, (3) describe what she **can afford** with a **budget line** whose slope is the **MRT** = p_F/p_E = how the market lets her trade the goods. The **optimum** is where the highest reachable indifference curve is **tangent** to the budget line — i.e. where **MRS** = **MRT** ($MU_F/MU_E = p_F/p_E$): the rate she's *willing* to trade equals the rate the market *lets* her trade.

The red thread. Every piece answers one of three questions: *what do I want?* (preferences → utility → MRS), *what can I afford?* (budget line → MRT), and *so what do I pick?* (tangency: MRS = MRT). Get those three and you can predict how choices shift when **prices** or **income** change — which is exactly a **demand curve** (next chapter). §5.6 then asks: *when does this rational machine break down?* (behavioral economics).

Key concepts / "modes" to use

- **Bundle** — a list of quantities of each good (e.g. "4 Entertainment, 2 Food"). Choice = ranking bundles.
- **Preferences & the rationality axioms** — **completeness** (can rank any two bundles) · **transitivity** (rankings "hold in transit") · **monotonicity** / "more is better" (convenient add-on).
- **Utility function** $U(F, E)$ — assigns a number to each bundle so that **higher** = **preferred**. **Ordinal**, not cardinal (only the *ranking* means anything).
- **Marginal utility (MU)** — extra utility from one more unit of a good, holding the other fixed = the **partial derivative**; typically **diminishing**.
- **Indifference curve** — all bundles giving the *same* utility; **5 properties**; three shapes (**perfect substitutes**, **imperfect substitutes**, **perfect complements**).
- **Marginal rate of substitution (MRS)** — slope of the **indifference curve** = MU_F / MU_E = how much E you'll give up for one more F at equal utility.

- **Budget constraint / line** — $I = p_F \cdot F + p_E \cdot E$; the **opportunity set** is everything on or under it.
- **Marginal rate of transformation (MRT)** — slope of the budget line = p_F / p_E = how the *market* lets you swap the goods.
- **Consumer optimum** — highest indifference curve tangent to the budget line: **MRS = MRT**, i.e. $MU_F / MU_E = p_F / p_E$ (equivalently, equal marginal utility *per euro*: $MU_F / p_F = MU_E / p_E$).
- **Interior vs corner solution** — tangency inside the axes vs "spend everything on one good" (perfect substitutes).
- **Behavioral economics** — systematic deviations: **social preferences, lack of self-control, cognitive biases** (loss aversion, endowment effect, framing, overconfidence).

Section-by-section main points

§5.1 How can we model choice? — the three ingredients (deck)

Core: we model an individual who **faces constraints** (income, time) and **does as well as possible** given them. Sounds almost tautological — the whole chapter is about *putting structure* on "does as well as possible." The deck lists the **ingredients we need**:

- A language to say "I like this more than that" → **preferences** (§5.2).
- **Assumptions** that make those preferences *rational* → the **axioms** (completeness, transitivity).
- A tool to quantify wellbeing → the **utility function** (§5.2.1).
- A tool to quantify "you can't have it all" → the **budget constraint** (§5.3).
- Bringing it together → **constrained optimization** (§5.4).
- Three premises of the whole build (deck + §5.1): (1) we know and have stable preferences, (2) we face constraints (time, money), (3) we try to do as well as possible given them.

§5.2 Preferences and the rationality axioms (deck slides 6–7)

Core: choice is modelled as **ranking bundles**. For the ranking to give a consistent model we assume it obeys two **essential** axioms (plus one convenient one). The deck's notation for the ranking:

Symbol	Reads	Meaning
$a > b$	"a is strictly preferred to b"	a is <i>better</i> than b
$a \geq b$	"a is weakly preferred to b"	a is <i>at least as good</i> as b
$a \sim b$	" indifferent between a and b"	a and b are <i>equally good</i>

- **Completeness (essential):** the consumer can **rank any two bundles** — for any A, B she can say $A > B$, $B > A$, or $A \sim B$. (No "I can't compare these.")
- **Transitivity (essential):** rankings "**hold in transit**" — if $A > B$ and $B > C$, then $A > C$. This is what lets us predict choice in a *chain* of comparisons (and stops preferences from cycling).
- **Monotonicity / "more is better" (convenient, not essential):** all else equal, **more of a good is better**. This is what makes bundles *farther from the origin* preferred. (Things you want *less of* — pollution, noise — are relabelled "*bads*," and we redefine the good as their *absence*.)

- **Why the axioms matter:** completeness + transitivity are *exactly* the conditions under which preferences can be represented by a utility function (next section). No axioms → no utility function → no model.

§5.2.1 Utility & utility functions (deck slides 8–9)

Core: if preferences are **complete and transitive**, we can represent them with a utility function $U(\cdot)$ that assigns each bundle a **number**, higher for more-preferred bundles. Neoclassical economics inherits this from the **utilitarian** tradition (J.S. Mill: "*actions are right in proportion as they tend to promote happiness*").

UTILITY REPRESENTS PREFERENCES

$$a \succ b \iff U(a) > U(b)$$

Reads: the utility function just **translates the ranking into real numbers** — bundle a is preferred to b if **and only if** a scores a higher utility number. (This is the **von Neumann–Morgenstern** representation the deck cites.) The function *orders* preferences; it does not measure "happiness" in any absolute unit.

- **Ordinal, not cardinal (the crucial caveat):** only the **ranking** matters, **not** the size of the gap. $U = 8$ vs $U = 4$ means the first is **preferred** — it does **not** mean "twice as good." (Deck: " $9 > 1$, but do '1' and '9' mean something meaningful?" — no. Utility is unobservable.)
- **General vs specific form:** general $U = U(F, E)$; a specific example the textbook uses is $U = \sqrt{EF}$. **Worked:** at $E=2, F=8$, $U = \sqrt{(2 \cdot 8)} = \sqrt{16} = 4$. Bundles with the **same product** $E \cdot F = 16$ (e.g. $E=4, F=4$ or $E=8, F=2$) sit on the **same** indifference curve; $E=5, F=5$ gives $U = 5 \rightarrow$ a **higher** curve.
- **Common functional forms (deck slide 12):**
 - **Cobb-Douglas:** $U = F^a \cdot E^b$ — the workhorse; convex, imperfect substitutes.
 - **Perfect substitutes:** $U = a \cdot F + b \cdot E$ — straight-line indifference curves.
 - (Perfect complements: $U = \min(F, E)$ — L-shaped.)

★ Marginal utility & diminishing marginal utility (deck slides 10, 16)

Core: marginal utility (MU) is the extra utility from **one more (infinitesimal) unit** of a good, holding the others fixed — i.e. the **partial derivative** of U . It is usually **diminishing** (the 1st slice of pizza beats the 7th).

MARGINAL UTILITY

$$MU_F = \partial U(F, E) / \partial F \quad \cdot \quad MU_E = \partial U(F, E) / \partial E$$

Reads: **how much total utility rises per extra unit of one good, other goods held constant** — $\partial U / \partial F$ = the partial derivative w.r.t. Food • **diminishing MU** = this derivative *shrinks* as you get more of the good.

- **The deck's three consumers (utility over one good):** for each extra apple —
 - **Bob:** each apple adds the **same** utility → **constant** MU (linear U).
 - **Linda:** each apple adds a bit **more** → **increasing** MU (convex U). (*unusual*)

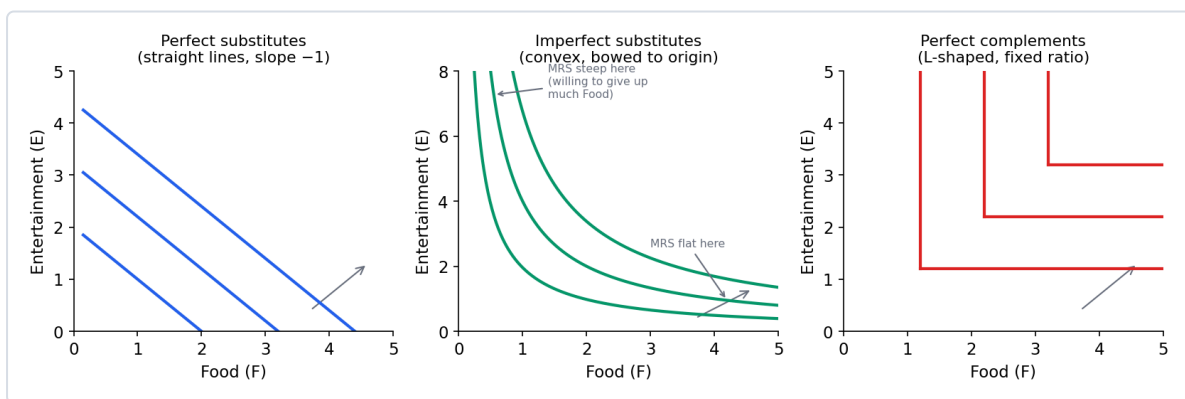
- **Mike:** each apple adds a bit **less** → **diminishing MU** (concave U). (*the normal case*)
- **Worked example** ($U = \sqrt{EF} = E^{0.5}F^{0.5}$): $MU_F = \partial U / \partial F = 0.5 \cdot E^{0.5} \cdot F^{-0.5} = 0.5 \cdot \sqrt{E/F}$. At $E=2, F=8$: $MU_F = 0.5 \cdot \sqrt{2/8} = 0.5 \cdot 0.5 = 0.25$. As F rises (holding E), $\sqrt{E/F}$ falls → MU_F falls → diminishing marginal utility. ✓
- **Why it matters:** diminishing MU is *why* indifference curves are **bowed toward the origin** (convex) and *why* the **MRS falls** as you move down a curve — the next two ideas.

Indifference curves — 5 properties & the 3 types (deck slides 13–14)

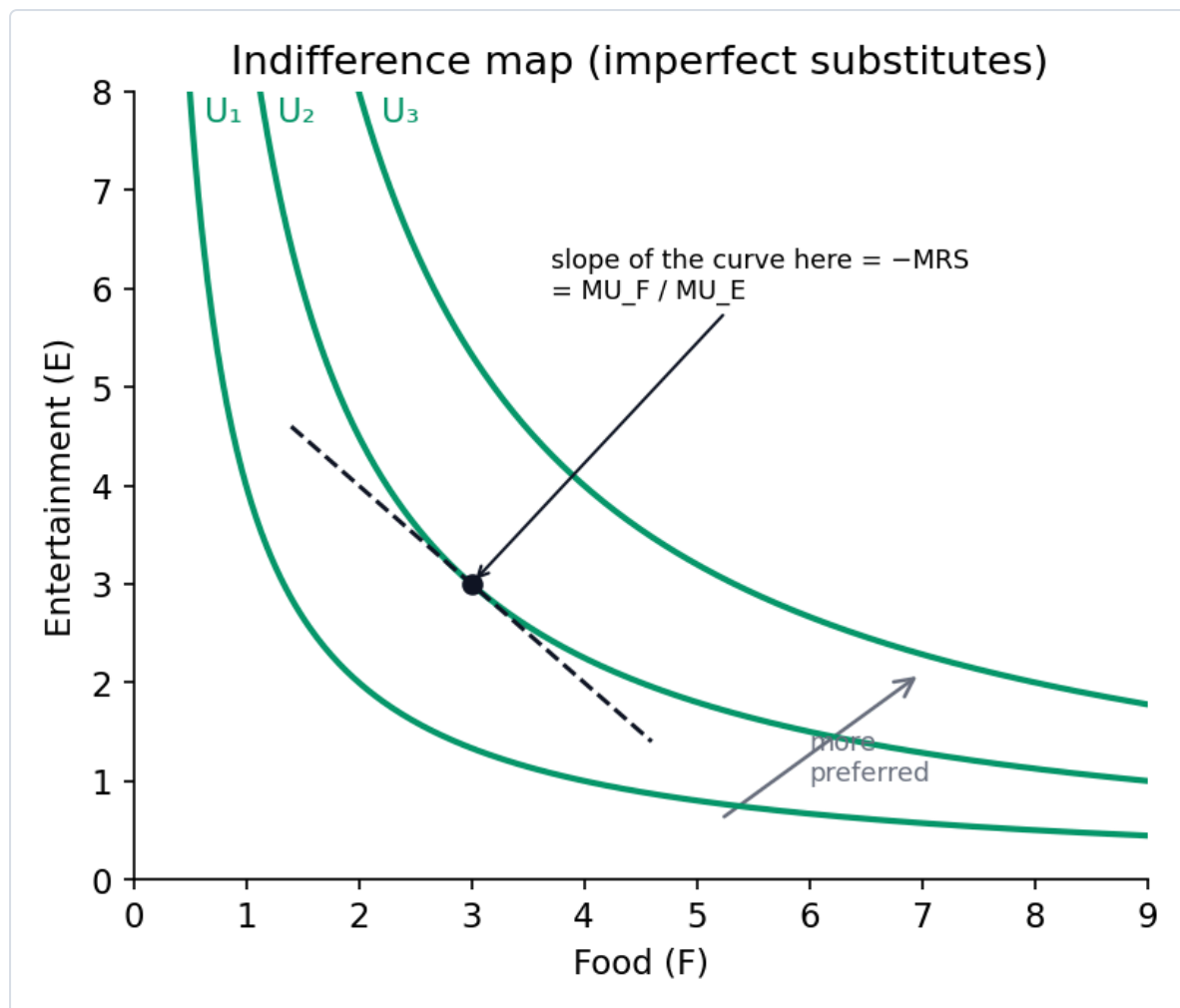
Core: an indifference curve connects all bundles the consumer views as equally desirable. It's the level-set of the utility function (fix U at one number). The deck's **five properties**:

#	Property	Why
1	Curves farther from the origin are preferred	"more is better" (monotonicity)
2	Every bundle lies on some <u>indifference curve</u>	completeness
3	Indifference curves cannot cross	transitivity (crossing → contradiction)
4	They slope downward	to keep utility fixed, more of one good ⇒ less of the other
5	They cannot be "thick"	a thick band would contain a strictly-better bundle (monotonicity)

- (Utility is read **ordinally**: the labels on the curves rank them; the gaps between labels are meaningless.)
- **Three shapes of preference** (textbook §5.2):



- **Perfect substitutes** — tradeable at a **constant rate** → **straight-line** curves (constant slope). *Example:* one €20 note ~four €5 notes, always. $U = aF + bE$.
- **Imperfect substitutes (the normal case)** — willing to trade, but the **rate depends on how much of each you already have** → **convex** curves, "bowed toward the origin" (strictly convex preferences). *When you have lots of chips and no drink, you'll give up many chips for one drink; when you have few chips, you won't.* $U = \sqrt{EF}$, Cobb-Douglas.
- **Perfect complements** — consumed in a **fixed ratio** → **L-shaped** curves. *Left & right shoes: a 2nd right shoe alone adds nothing.* $U = \min(F, E)$.



How to read it (indifference map). Each green curve is one utility level; **northeast = more preferred** (property 1). The dashed tangent at the marked bundle has slope $-MRS$: it gets **flatter** as you move right (more Food, less willing to give up Food) — that's diminishing MRS, the topic below.

★ Marginal rate of substitution (MRS) (deck slides 17–19)

Core: the **MRS** is the slope of the indifference curve — the maximum amount of one good the consumer will give up to get one more unit of the other, staying equally happy. Because utility is constant along the curve ($dU = 0$), it equals the **ratio of marginal utilities**.

MARGINAL RATE OF SUBSTITUTION

$$MRS = -dE/dF = MU_F / MU_E \quad (\text{good on the x-axis over good on the y-axis})$$

Reads: how much E you'll trade for one more F and stay on the same curve — derive it by **totally differentiating** U and setting $dU = 0$: $0 = MU_F \cdot dF + MU_E \cdot dE \rightarrow -dE/dF = MU_F / MU_E$. MU_F = marginal utility of the x-axis good (Food) • MU_E = marginal utility of the y-axis good (Entertainment) • the ratio = the curve's steepness at that point.

- **Diminishing MRS (why the curve bows):** as you move **down-right** along the curve (more F, less E), MU_F falls and MU_E rises, so the ratio MU_F / MU_E **shrinks** — the curve **flattens**. "When I have relatively less pizza I'll give up more burritos for a slice than when I already have lots of pizza."

- **Worked example ($U = \sqrt{EF}$):** $MU_F = 0.5\sqrt{E/F}$, $MU_E = 0.5\sqrt{F/E}$, so $MRS = MU_F/MU_E = (0.5\sqrt{E/F})/(0.5\sqrt{F/E}) = E/F$. At bundle $E=2, F=8$: $MRS = 2/8 = 0.25 \rightarrow$ she'll give up only **0.25 units of E** for one more F (she already has lots of F). At $E=8, F=2$: $MRS = 8/2 = 4 \rightarrow$ now she'll give up **4 units of E** for one more F. *Same curve, opposite ends $\rightarrow$ diminishing MRS.* ✓

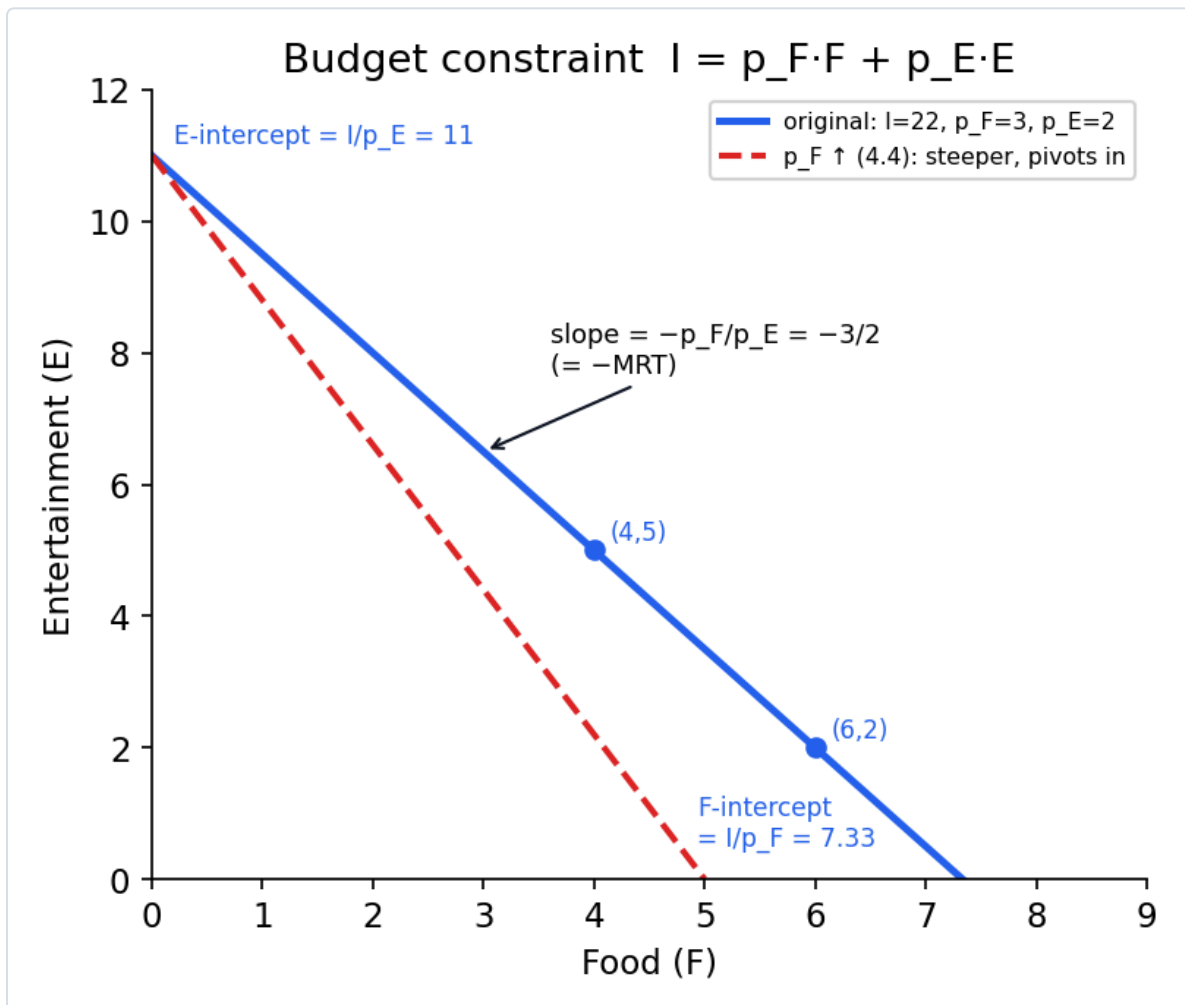
§5.3 Budget constraint & the MRT (deck slides 21–22)

Core: the **budget constraint** operationalizes "you can't have it all": given prices and income I , total spending can't exceed income. With two goods it's a **straight line** (the **budget line**); everything on or below it is the **opportunity set**.

BUDGET CONSTRAINT & LINE

$$I = p_F F + p_E E \implies E = I/p_E - (p_F/p_E) \cdot F$$

Reads: income = spending on Food + spending on Entertainment; rearranged for graphing, E-intercept = I/p_E (all income on E), slope = $-p_F/p_E$, F-intercept = I/p_F (all income on F).



- **Worked example (deck: $I=22, p_F=3, p_E=2$):** E-intercept = $22/2 = 11$; slope = $-3/2$; F-intercept = $22/3 \approx 7.33$. Bundle $(F=4, E=5)$ is affordable: $3 \cdot 4 + 2 \cdot 5 = 22$ ✓. So is $(F=6, E=2)$: $3 \cdot 6 + 2 \cdot 2 = 22$ ✓.

- **MRT — marginal rate of transformation** = the **slope of the budget line** = how the **market** lets you swap goods:

MARGINAL RATE OF TRANSFORMATION

$$MRT = p_F / p_E \quad (\text{the relative price = the market "exchange rate" between the goods})$$

Reads: give up one F, the market gives you p_F/p_E units of E. With $p_F=3$, $p_E=2$, $MRT = 1.5$: one less Food frees €3, which buys 1.5 Entertainment. (*MRS = what you're willing to trade; MRT = what the market lets you trade.*)

- **Comparative statics of the line (deck):**
 - **Income change** → **parallel shift** (slope unchanged; both intercepts move).
 - **Price change** → **pivot** (steepness changes). A higher $p_F \Rightarrow$ **steeper**, pivoting **inward** around the E-intercept (see red dashed line); a lower $p_F \Rightarrow$ **flatter**.

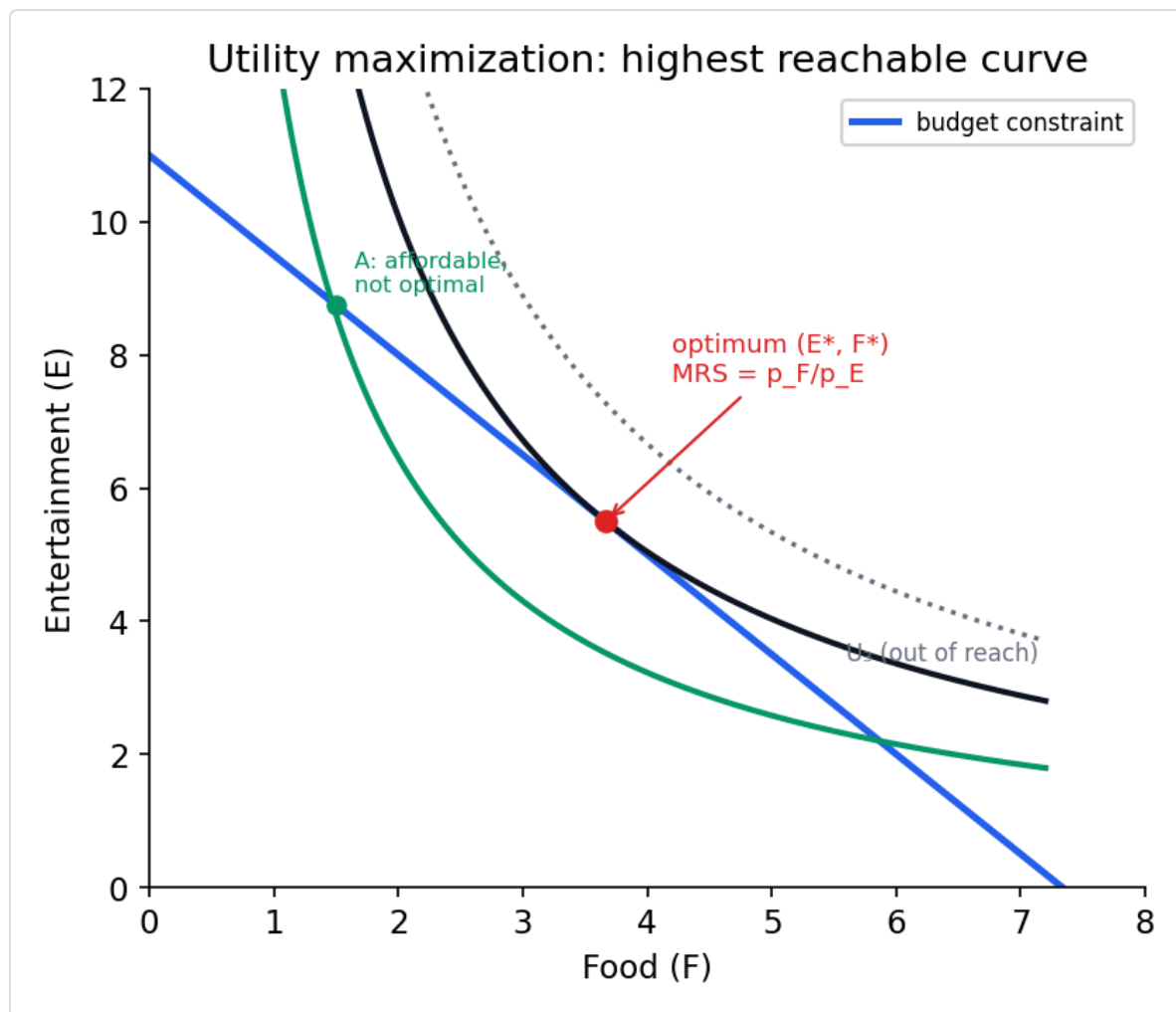
★ §5.4 The optimum: maximize utility subject to the budget (deck slide 23)

Core: the consumer solves **max $U(F,E)$ subject to $p_F \cdot F + p_E \cdot E \leq I$** . Graphically: find the **highest indifference curve** that still touches the budget line — that's the **tangency point**. At the tangency, the slope of the indifference curve (**MRS**) equals the slope of the budget line (**MRT**).

CONSUMER OPTIMUM (TANGENCY)

$$MRS = MRT \iff MU_F/MU_E = p_F/p_E \iff MU_F/p_F = MU_E/p_E$$

Reads: at the optimum, **the rate you're willing to trade = the rate the market lets you trade**. The last form is the "**equal marginal utility per euro**" (bang-for-the-buck) rule: **the last euro spent on Food yields the same extra utility as the last euro spent on Entertainment** — otherwise you'd reallocate toward the higher-MU-per-euro good.



How to read it (utility maximization). Point A is *affordable* (on the budget line) but **not optimal**: the curve through A is steeper than the line ($MRS > MRT$), so she values Food more than the market charges → she should **buy more Food, less Entertainment**, sliding down to the tangency. The **red optimum** is the best reachable bundle; the dotted curve U_3 would be better but her income can't reach it.

- **Why A isn't optimal (the deck's logic):** at A, $MU_F/MU_E > p_F/p_E$ — food is "cheap relative to how much you value it." Reallocate toward food until the two ratios are equal.
- **Worked example ($U = \sqrt{EF}$), $I=22$, $p_F=3$, $p_E=2$):** $MRS = E/F$. Set $MRS = MRT$: $E/F = 3/2$ → $E = 1.5F$. Plug into the budget $3F + 2E = 22$: $3F + 2(1.5F) = 22$ → $6F = 22$ → $F^* = 3.67$, $E^* = 5.5$. (Check: $3 \cdot 3.67 + 2 \cdot 5.5 = 11 + 11 = 22$ ✓ — matches the red dot in the figure.)

Comparative statics · interior vs corner · the Cobb-Douglas shortcut (deck slide 24, §5.7)

Core: once you can find the optimum, you can ask **how it moves** when a price or income changes — and that movement, traced out, **is the demand curve** (Ch. 6). Watch for **corner solutions**.

- **Interior vs corner solution (deck):**
 - **Interior solution** — the tangency lies **strictly inside** the axes (buy *some* of both goods). Requires **convex** (imperfect-substitute) preferences.
 - **Corner solution** — the optimum is **on an axis** (buy *only one* good). Happens with **perfect substitutes**: if $MRS \neq MRT$ everywhere, you spend **everything** on whichever good gives more

utility per euro (MU/p). Example: if $U = 2F + E$ and $p_F = p_E$, Food gives twice the bang per euro $\rightarrow$ buy only Food.

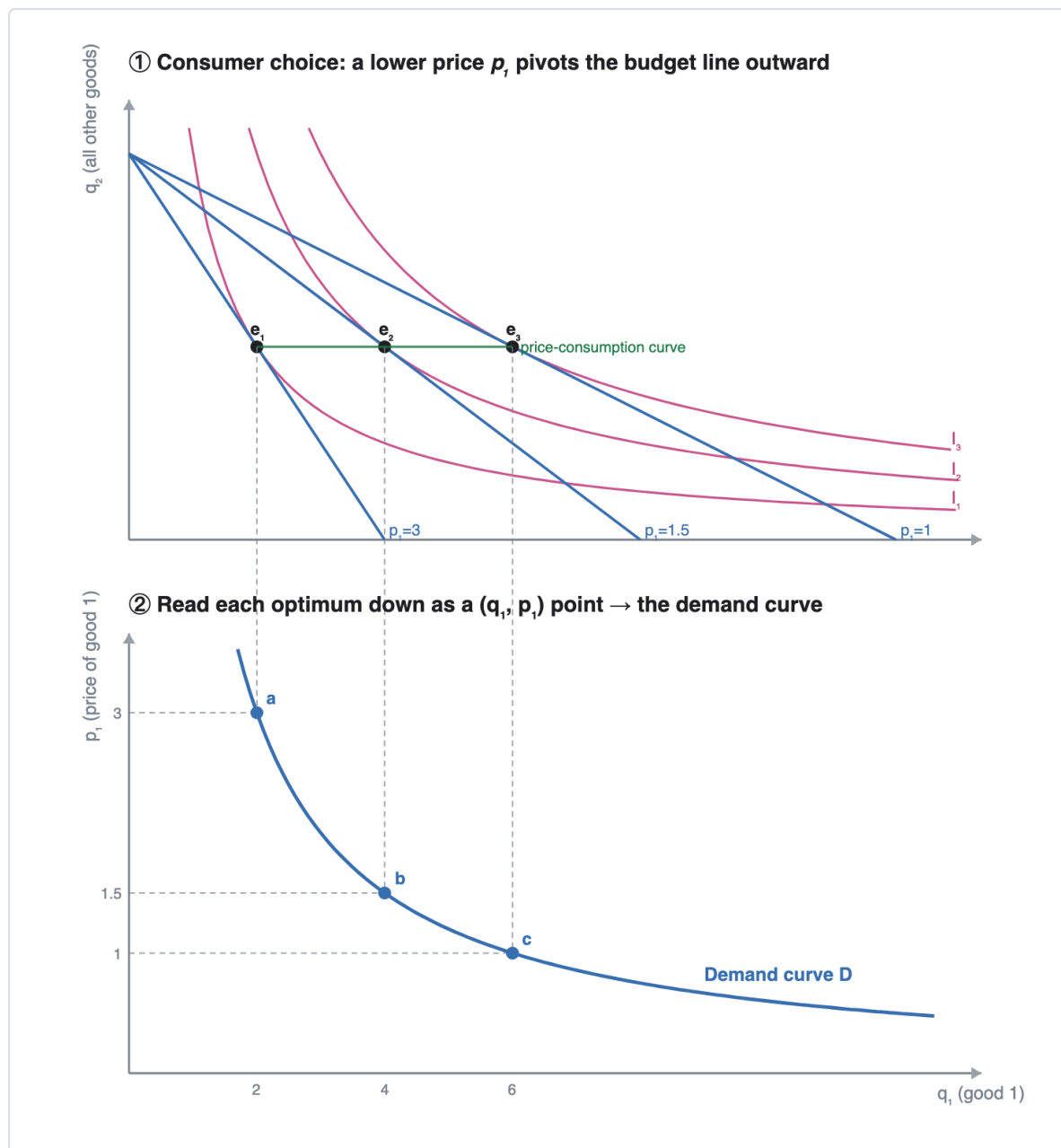
- **Perfect complements** — always a corner-of-the-L: buy the goods in their **fixed ratio** regardless of the price ratio (solve the ratio + the budget line together).
- **The Cobb-Douglas shortcut (worth memorising for exams):** for $U = F^a \cdot E^b$, the tangency condition solves in closed form — the consumer **spends a constant share of income on each good**:

COBB-DOUGLAS OPTIMAL DEMANDS

$$F^* = \left[\frac{a}{a+b} \right] \cdot I/p_F \quad \cdot \quad E^* = \left[\frac{b}{a+b} \right] \cdot I/p_E$$

Reads: spend the fraction $a/(a+b)$ of income on Food and $b/(a+b)$ on Entertainment, then divide by each price to get quantities. The exponents are the expenditure shares. Worked ($U = F^{0.5}E^{0.5}$), so $a=b=0.5$, $I=22$, $p_F=3$, $p_E=2$): $F^ = 0.5 \cdot 22/3 = 3.67$, $E^* = 0.5 \cdot 22/2 = 5.5$ — same answer as the tangency method above, in one line. (Notice each good's demand **doesn't depend on the other good's price** — a special Cobb-Douglas feature.)*

- **The appendix method (§5.7) — Lagrange:** for a general U , combine objective + constraint into $\mathcal{L} = U(F,E) + \lambda(I - p_F F - p_E E)$, set the three partials to zero, and the first two give back $MU_F/MU_E = p_F/p_E$ — the same tangency condition. λ = the **marginal utility of income** ("how much extra utility one more euro of income buys"). *You won't need Lagrange for the exam, but know that it reproduces $MRS = MRT$.*
- **The payoff (Ch. 6 foreshadow):** change p_F , re-solve the optimum, and record the new F^* — repeat, and the (p_F, F^*) pairs **trace out the demand curve**. *This is why we built all this machinery: choice $\rightarrow$ demand.*



How to read it (choice → demand, step by step). This is the picture behind "trace out the demand curve." ① (top) each time the price of the good **falls**, the budget line **pivots outward** and the tangency optimum slides to **more of the good** ($e_1 \rightarrow e_2 \rightarrow e_3$). ② (bottom) drop each optimum straight down to plot its (quantity, price) pair — **a, b, c** — and **join them: that's the demand curve**. The demand curve you took as given in [Lecture 2](#) is nothing but the consumer's optimum **re-solved for every price**. (Fully worked in [Lecture 5 — Deriving the demand curve](#); and see [practice Q12](#), which computes two of these points, ($p=4, F=5$) and ($p=2, F=10$).)

§5.5–5.6 Discussion of the framework & behavioral economics

Core: the rational-choice model is powerful (Becker: "maximizing behavior, market equilibrium, and stable preferences, used relentlessly and unflinchingly") — but it's a **model**. Two clarifications, then the **deviations**.

• Two common misconceptions (§5.5):

- "Economics says be a selfish materialist." **No** — the framework maximizes **utility**, not **goods**. Giving to charity, choosing a low-paid academic job you love, buying organic — all perfectly rational if that's

your preference.

- "It only works for apples-vs-oranges." Actually **Gary Becker** extended it to fertility, marriage, crime, discrimination: assume **stable preferences** and let **costs/benefits differ** across contexts (e.g. why fertility differs across countries — Table 5.2 — need not be different *tastes*).

- **Behavioral economics (§5.6)** — three deviations from the simple model:

Deviation	What it is	Signature examples
Social preferences	We care about others — fairness, status, altruism, spite. <i>Not irrational</i> — just richer preferences.	Ultimatum game : proposers offer ~40% (not ~0%), responders reject ~16% of low offers to punish unfairness.
Lack of self-control	We over-weight the present — a myopic " <i>doer</i> " vs a " <i>planner</i> " (Thaler–Shefrin).	Smoking, over-spending, under-saving; the marshmallow test; policy responses = mandatory pensions, alcohol limits.
Cognitive limitations & biases	Systematic (not random) errors; bounded rationality , heuristics.	Loss aversion / status-quo bias / endowment effect (Cornell coffee-mug study: owners value the mug 2–4× more); framing (600-lives problem); overconfidence (68% of entrepreneurs think they'll beat the odds).

- **Random error vs bias (deck's dartboard)**: *random* misses scatter around the bull's-eye (people are right *on average*); a **behavioral bias** is a **systematic** skew off-centre (Kahneman & Tversky, *Thinking, Fast and Slow*).
- **Final word (§5.6.5)**: the rational model stays the **benchmark** — you need it *even to define* a "deviation." Behavioral economics tells you **when** the benchmark will mislead (e.g. demand for addictive painkillers vs baking soda).

Cases & examples

- **Why did shrimp consumption triple? (Freakonomics / Levitt)** — §5.4.1. Non-economists guessed **demand shifts** (health fads, movies, trendy chefs). The choice framework points first to **prices and income**: the **relative price of shrimp fell** — that alone moves the optimum along/across the budget line. *Lesson: shifts in the budget constraint, not just tastes, drive choices.*
- **Same preferences, different choices: US pizza vs French duck confit (Dubois et al. 2014)** — Fig 5.8. Give an American and a French consumer the **identical** indifference map. Pizza is **cheaper in the US**, confit **cheaper in France** → different **budget-line slopes** → the American picks more pizza, the French more confit. *Different choices need not mean different tastes — just different relative prices.*
- **The ultimatum game — fairness is real (Güth 1982; Henrich 2001)**. Split €10; responder can veto (then both get €0). Pure self-interest predicts "offer €0.10, accept." Instead proposers offer ~40%, and low offers get **rejected** — even across 15 small-scale societies worldwide (offers never below ~25%). *Evidence for social preferences.*
- **The Cornell coffee-mug experiment — the endowment effect (Kahneman et al. 1990)**. Mugs handed out at random; owners then valued them 2–4× **higher** than non-owners would pay. Ownership shouldn't matter to a rational valuation — but a **reference point** does. *Loss aversion / status-quo bias in one image.*

- **Food stamps or cash? (deck slide 25).** ~50% of US kids use food stamps before age 20. In the standard model, **cash is weakly better** (it relaxes the budget line *more* — you can buy food *or* anything else), so an in-kind transfer can only tie or lose. *Yet* food stamps persist — because of **paternalism** / **self-control** concerns and political constraints. *A perfect budget-constraint application with a behavioral twist.*
- **Fertility rates: tastes or costs? (Becker; Table 5.2).** South Korea 0.97 kids/woman vs Mali 6.14. A Becker-style reading: **same preferences**, different **costs/benefits** of children (childcare compatibility with careers, old-age support, child mortality, pensions). *Rational choice reaches far beyond "goods."*

Formula sheet (Consumer Choice)

Concept	Formula	In words
Utility represents preferences	$a > b \Leftrightarrow U(a) > U(b)$	numbers rank bundles (ordinal)
Marginal utility	$MU_F = \partial U / \partial F$	extra utility per extra unit; usually diminishing
MRS (slope of indifference curve)	$MRS = MU_F / MU_E$	how much E you'll trade for one more F, same utility
Budget line	$I = p_F \cdot F + p_E \cdot E \rightarrow E = I/p_E - (p_F/p_E)F$	what you can afford; slope $-p_F/p_E$
MRT (slope of budget line)	$MRT = p_F / p_E$	how the market lets you trade F for E
Consumer optimum	$MRS = MRT \Leftrightarrow MU_F/MU_E = p_F/p_E \Leftrightarrow MU_F/p_F = MU_E/p_E$	tangency = equal MU per euro
Cobb-Douglas demands ($U = F^a E^b$)	$F^* = a/(a+b) \cdot I/p_F$, $E^* = b/(a+b) \cdot I/p_E$	spend constant income shares

Key terms

Term	One-line meaning
Bundle	a specific quantity of each good
Completeness	can rank <i>any</i> two bundles
Transitivity	$A > B, B > C \Rightarrow A > C$ (rankings "hold in transit")
Monotonicity	"more is better" — farther-out curves preferred
Utility function	numbers on bundles preserving the ranking (ordinal)
Marginal utility (MU)	$\partial U / \partial(\text{good})$; extra utility from one more unit
Indifference curve	all bundles of equal utility
MRS	slope of indifference curve = MU_F / MU_E
Budget constraint / line	$I = p_F F + p_E E$; the opportunity set
MRT	slope of budget line = p_F / p_E
Consumer optimum	highest reachable curve tangent to the budget line
Interior / corner solution	some of both goods / all of one good
Behavioral economics	systematic deviations from the rational model
Loss aversion / endowment effect	losses hurt more than equal gains; owning raises valuation

Exam pointers

- **Draw the machine, then solve it.** For "find the optimal bundle": (1) write $MRS = MU_F / MU_E$, (2) set $MRS = p_F / p_E$, (3) combine with the **budget line** $I = p_F F + p_E E$, (4) solve the two equations. *Two equations, two unknowns — always.*
- **Cobb-Douglas → use the shortcut.** If $U = F^a E^b$, jump straight to $F^* = a / (a+b) \cdot I / p_F$. Saves time; derive it once so you trust it.
- **Check the second good.** After finding one quantity, plug back into the budget line for the other, and **verify it satisfies the constraint** ($p_F F^* + p_E E^* = I$).
- **MRS = MU of the x-axis good over MU of the y-axis good.** Getting the ratio upside-down is the #1 error. If Food is on the x-axis, $MRS = MU_F / MU_E$ and the tangency is $MU_F / MU_E = p_F / p_E$.
- **Corner solutions:** perfect substitutes → compare MU_F / p_F vs MU_E / p_E and spend everything on the winner. Perfect complements → ignore the price ratio, use the **fixed ratio + budget line**.
- **Comparative statics language:** income change = **parallel shift**; own-price change = **pivot**. Re-solving the optimum after a price change is how you **derive a demand curve** (Ch. 6).
- **Ordinal, not cardinal:** never say "twice as happy." Utility numbers only **rank**.
- **Behavioral:** be able to name the bias from a scenario (endowment effect, loss aversion, framing, self-control, overconfidence) and say it's a **systematic** (not random) deviation.

Bridge to Lecture 5 (Ch. 6 — Demand curves & elasticities). We now have a consumer who picks the tangency bundle for *given* prices and income. **Change one price and re-solve**, over and over, and the optimal quantities trace out a **demand curve** — closing the loop back to the demand we assumed in Lecture 2, and connecting to the **elasticities** of Lecture 3.

★ Practice problems (with solutions)

A full-chapter set: preferences & axioms → utility & MU → MRS → budget line & MRT → the optimum → Cobb-Douglas shortcut → corner solutions → comparative statics → behavioral. Try each before opening the solution. Use **Food (F)** on the x-axis, **Entertainment (E)** on the y-axis throughout.

1. A consumer has bundles $A = (E\ 4, F\ 2)$, $B = (E\ 8, F\ 2)$, $C = (E\ 3, F\ 3)$, $D = (E\ 6, F\ 5)$. Using only "**more is better**," which rankings can you establish for certain, and which need the indifference map? (Compare A–B, C–D, and A–C.)
2. Preferences satisfy completeness, transitivity, monotonicity. A student says: "*I prefer coffee to tea, tea to cocoa, but cocoa to coffee.*" Which **axiom** does this violate, and why is it a problem for building a utility function?
3. $U = \sqrt{EF}$. (a) Find the utility of $(E\ 2, F\ 8)$, $(E\ 4, F\ 4)$, $(E\ 8, F\ 2)$, $(E\ 5, F\ 5)$. (b) Which lie on the same indifference curve? (c) Rank all four bundles.
4. $U = F^{0.5} E^{0.5}$. (a) Derive MU_F and MU_E . (b) Evaluate both at $(F\ 4, E\ 9)$. (c) Does MU_F rise or fall as F increases (E fixed)? Name the principle.
5. For $U = \sqrt{EF}$, show that $MRS = E/F$. Then compute the MRS at $(E\ 2, F\ 8)$ and at $(E\ 8, F\ 2)$ and interpret the difference in one sentence.
6. Income $I = 40$, $p_F = 4$, $p_E = 2$. (a) Write the budget line as $E = \dots$. (b) Give both intercepts and the slope. (c) Is $(F\ 6, E\ 8)$ affordable? (d) What is the **MRT**, and what does it mean?
7. Same as 6 ($I = 40$, $p_F = 4$, $p_E = 2$). (a) The government gives a **€20 income top-up**. What happens to the budget line? (b) Instead, p_F falls to 2. What happens? (c) Sketch the difference in words (shift vs pivot).
8. $U = \sqrt{EF}$, $I = 40$, $p_F = 4$, $p_E = 2$. Find the **optimal bundle** using $MRS = MRT$ + the budget line. Verify it satisfies the constraint.
9. $U = F^{0.25} E^{0.75}$ (Cobb-Douglas), $I = 60$, $p_F = 3$, $p_E = 6$. Use the **expenditure-share shortcut** to find F^* and E^* . What share of income goes to each good?
10. (Perfect substitutes.) $U = 2F + E$, $I = 24$, $p_F = 3$, $p_E = 2$. (a) Compute MU_F/p_F and MU_E/p_E . (b) What is the optimal bundle, and why is it a **corner solution**?
11. (Perfect complements.) $U = \min(F, E)$, $I = 30$, $p_F = 2$, $p_E = 3$. Find the optimal bundle. (Hint: the optimum sits at the corner where $F = E$; combine with the budget line.)
12. (Comparative statics.) Start from problem 8's optimum ($U = \sqrt{EF}$, $I=40$, $p_F=4$, $p_E=2$). Now p_F falls to €2. (a) Find the new optimum. (b) Did the quantity of Food rise or fall? (c) You now have **two (p_F, F^*) points** — what curve are you tracing out?
13. (Behavioral.) Name the bias in each: (a) You keep a concert ticket you'd never pay today's resale price for. (b) "1 in 3 patients survive" feels better than "2 in 3 die." (c) You put your alarm clock across the

room so you can't hit snooze. (d) 80% of new founders rate their success odds $\geq 7/10$.

Solutions

1. **A vs B:** B has more E (8 vs 4), same F $\rightarrow B > A$ (more is better). **C vs D:** D has more of *both* (E $6 > 3$, F $5 > 3$) $\rightarrow D > C$. **A vs C:** A has more E ($4 > 3$) but less F ($2 < 3$) $\rightarrow$ **cannot rank** by "more is better" alone; you need the **indifference map** (depends on the MRS).
2. Violates **transitivity**. coffee > tea and tea > cocoa should imply coffee > cocoa, but she says cocoa > coffee — the preferences **cycle**. A cycling ranking **can't be represented by a utility function** (no set of numbers can have coffee > tea > cocoa > coffee), so the whole model of choice breaks down.
3. (a) $U = \sqrt{EF}$: (E2,F8) $\rightarrow \sqrt{16} = 4$; (E4,F4) $\rightarrow \sqrt{16} = 4$; (E8,F2) $\rightarrow \sqrt{16} = 4$; (E5,F5) $\rightarrow \sqrt{25} = 5$. (b) The **first three** (all $E \cdot F = 16$) lie on the **same indifference curve** ($U=4$). (c) **(E5,F5) > {the other three, which are indifferent to each other}** — i.e. $(5,5) > (2,8) \sim (4,4) \sim (8,2)$.
4. (a) $MU_F = \partial U / \partial F = 0.5 F^{-0.5} E^{0.5} = 0.5 \sqrt{E/F}$; $MU_E = 0.5 F^{0.5} E^{-0.5} = 0.5 \sqrt{F/E}$. (b) At (F4,E9): $MU_F = 0.5 \sqrt{9/4} = 0.5 \cdot 1.5 = 0.75$; $MU_E = 0.5 \sqrt{4/9} = 0.5 \cdot 0.667 = 0.333$. (c) As F rises, $\sqrt{E/F}$ falls $\rightarrow MU_F$ falls = **diminishing marginal utility**.
5. $MRS = MU_F / MU_E = [0.5 \sqrt{E/F}] / [0.5 \sqrt{F/E}] = \sqrt{E/F} \cdot \sqrt{E/F} = E/F$. At (E2,F8): $MRS = 2/8 = 0.25$; at (E8,F2): $MRS = 8/2 = 4$. **Interpretation:** with lots of F she'll give up only 0.25 E for one more F; with little F she'll give up 4 E — the MRS **falls** as F rises (diminishing MRS / convex curve).
6. (a) $40 = 4F + 2E \rightarrow E = 20 - 2F$. (b) E-intercept = 20, F-intercept = 10, slope = -2. (c) (F6,E8): $4 \cdot 6 + 2 \cdot 8 = 24 + 16 = 40 \rightarrow$ **exactly affordable** (on the line). (d) $MRT = p_F / p_E = 4/2 = 2 \rightarrow$ giving up 1 Food frees €4, which buys 2 Entertainment.
7. (a) **Income +€20** ($I = 60$): **parallel outward shift** — new intercepts $E = 60/2 = 30$, $F = 60/4 = 15$; **slope unchanged** (-2). (b) **p_F falls to 2:** budget **pivots** — E-intercept **unchanged at 20**, F-intercept rises to $40/2 = 20$, slope flattens to $-p_F / p_E = -1$. (c) Income change = **parallel shift** (same slope, more of everything); own-price change = **pivot** (one intercept fixed, slope changes).
8. $MRS = E/F$ (from problem 5). Set $MRS = MRT$: $E/F = 4/2 = 2 \rightarrow E = 2F$. Budget: $4F + 2E = 40 \rightarrow 4F + 2(2F) = 40 \rightarrow 8F = 40 \rightarrow F^* = 5, E^* = 10$. (Check: $4 \cdot 5 + 2 \cdot 10 = 20 + 20 = 40$ ✓.)
9. Shortcut for $U = F^{0.25} E^{0.75}$ ($a=0.25, b=0.75, a+b=1$): $F^* = 0.25 \cdot 60/3 = 0.25 \cdot 20 = 5$; $E^* = 0.75 \cdot 60/6 = 0.75 \cdot 10 = 7.5$. **Shares:** 25% of income (€15) on Food, 75% (€45) on Entertainment. (Check: $3 \cdot 5 + 6 \cdot 7.5 = 15 + 45 = 60$ ✓.)
10. $U = 2F + E$: $MU_F = 2$, $MU_E = 1$. (a) $MU_F / p_F = 2/3 \approx 0.67$; $MU_E / p_E = 1/2 = 0.5$. (b) Food gives **more utility per euro** ($0.67 > 0.5$) and the rate is **constant** (perfect substitutes), so spend **everything on Food**: $F^* = I / p_F = 24/3 = 8$, $E^* = 0$. **Corner solution** because $MU_F / p_F \neq MU_E / p_E$ everywhere — there's no interior tangency; the highest indifference (straight) line touches the budget line **on the F-axis**.
11. $U = \min(F, E)$: no point in having more of one than the other, so the optimum has **$F = E$** . Budget: $2F + 3E = 30$ with $E = F \rightarrow 2F + 3F = 30 \rightarrow 5F = 30 \rightarrow F^* = E^* = 6$. (Check: $2 \cdot 6 + 3 \cdot 6 = 12 + 18 = 30$ ✓.) (Note: the price ratio is irrelevant to the ratio of goods — only to how much of the fixed bundle you can afford.)

12. Now $p_F = 2$, $p_E = 2$, $I = 40$. $MRS = E/F = p_F/p_E = 2/2 = 1 \rightarrow E = F$. Budget: $2F + 2E = 40$ with $E = F \rightarrow 4F = 40 \rightarrow F^* = 10, E^* = 10$. (b) Food rose from 5 to 10 as its price fell $4 \rightarrow 2$ (a downward-sloping demand response). (c) The two points ($p_F=4, F=5$) and ($p_F=2, F=10$) are two points on the consumer's demand curve for Food.

13. (a) **Endowment effect / sunk-cost + status-quo bias** (you value it more because you own it). (b) **Framing effect** (same facts, "survive" vs "die"). (c) **Lack of self-control** — a commitment device (planner constrains the future doer). (d) **Overconfidence**.

★ Fun facts & memorable details (Lecture 4)

Sticky bits from Consumer Choice.

- **"More is better" has a loophole:** economists just **redefine bads as the absence of the bad** — you don't demand pollution, you demand *clean air*. A neat trick that keeps monotonicity intact.
- **Utility is unobservable — so numbers lie.** $U = 8$ is not "twice as good" as $U = 4$; utility is purely **ordinal**. Early economists agonised over this; the modern answer is "we only ever need the ranking."
- **The bang-for-your-buck rule is the whole optimum.** $MU_F/p_F = MU_E/p_E$ — at the best bundle, the **last euro** spent on each good buys the **same** extra utility. If it didn't, you'd move money to the better-value good. That's *all* tangency means.
- **Cobb-Douglas people are creatures of habit:** with $U = F^a E^b$ you always spend the **same share** of income on each good — no matter how rich you get or how prices move. That's why it's the field's favourite functional form.
- **Left shoes are worthless alone.** Perfect complements (L-shaped curves) are why a single earring, one glove, or a lone right shoe adds **zero** utility — you need the matching unit.
- **The ultimatum game breaks "Homo economicus."** Offered a lopsided split, people **pay real money to punish unfairness** — and this holds from Pittsburgh to Papua New Guinea. Fairness is not a Western quirk.
- **You'd sell your coffee mug for far more than you'd pay for it.** The **endowment effect**: Cornell students valued a randomly-given mug **2–4× higher** than students who didn't get one. Ownership rewires valuation.
- **The paradox of voting:** a strict cost-benefit reading says *don't vote* (your vote almost never decides the election). That people vote anyway is a clue that **narrow self-interest is too small a theory of us**.
- **Ulysses tied himself to the mast** to resist the Sirens — a 3,000-year-old **commitment device**, the same logic as not keeping cigarettes in the house. Self-control problems are ancient.